

Talaix — Earth-Observation Data Requirements

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1. Purpose

This document answers one question precisely: what information does Talaix need from satellites. The platform is live today and already runs on open Earth-observation (EO) data. Section 2 lists what is integrated in production; Section 3 lists exactly what we need next — sensor, resolution, revisit and the reason for each item.

2. What the platform uses today (integrated, in production)

Variable	Mission / sensor	Resolution	Revisit / latency	Used for
Vegetation & fuel state, burn scars	Sentinel-2 L2A	10–20 m	5 days	wildfire, drought, land-use
Optical fallback archive	Landsat 8/9 Coll. 2 L2	30 m	8 days combined	gap filling, history
Active fires (NRT)	VIIRS S-NPP (NASA FIRMS)	375 m	~3 h latency	wildfire detection & alerts
Active fires (NRT)	MODIS (NASA FIRMS)	1 km	~3 h latency	wildfire detection
Land cover	ESA WorldCover v200	10 m	annual	exposure & fuel models
Elevation (Europe)	EU-DEM (Copernicus/EEA)	25 m	static	flood, slope, fire spread
Elevation (global)	SRTM (NASA)	90 m	static	same, outside Europe
Fire weather & climate history	ERA5 / ERA5-Land (C3S)	25 / 11 km	hourly, 1940 to now	FWI, drought, heat, wind
Soil moisture 0–7 cm	ERA5-Land (C3S)	11 km	hourly	drought
River discharge	GloFAS (EMS/JRC) + GEOGLOWS	~10 km / river reaches	daily forecasts	flood
Aerosols / dust	CAMS (ECMWF)	10–40 km	forecasts	dust & smoke exposure
Cyclone tracks	IBTrACS + GDACS (JRC)	storm fixes	3-hourly	tropical cyclones

Note: ERA5, GloFAS and CAMS are model/reanalysis products fed by satellite and in-situ observations — EO-derived data. Together with official non-satellite sources (Open-Meteo, USGS, WorldPop, OpenStreetMap), the platform integrates ~25 datasets selected from a 168-source audited observatory.

3. What we need next — the precise requirements

Priority	Requirement	Mission / sensor	Resolution	Revisit	Why
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P1	Flood extent through clouds, all-weather	Sentinel-1 SAR (IW, VV/VH)	10 m	6 days	flood mapping & damage evidence — our #1 gap
P2	Better terrain for flood & slope	Copernicus DEM GLO-30	30 m	static	replaces 90 m SRTM outside Europe
P3	Land-surface temperature for heat	Landsat 8/9 TIRS; later ESA LSTM	100 m	8–16 days	urban-heat evidence at asset level
P4	Daily coarse fire & temperature coverage	Sentinel-3 SLSTR	500 m–1 km	daily	fills VIIRS gaps, fire radiative power
P5	Smoke & dust exposure (NO ₂ , aerosol)	Sentinel-5P TROPOMI	~5 km	daily	health-relevant exposure evidence
P6	Burned-area history (training labels)	MODIS MCD64A1	500 m	monthly	fire-severity validation
P7	Very-high-resolution optical for asset validation	Pléiades / Pléiades Neo (Airbus, via ESA TPM)	0.3–0.5 m	on demand	verifies site findings for paying customers
P8	NRT SAR tasking during flood events	ICEYE or equivalent (commercial)	1–3 m	hours	event-response evidence while floods unfold

P1–P6 are open Copernicus/NASA products — the work is integration, plus guidance on doing it right. P7–P8 are commercial; we need them only for paid validation cases, and they are unaffordable for a bootstrapped pre-company project without an access scheme.

4. Latency and refresh requirements by product

Product	Refresh needed	Latency tolerance
Compliance evidence packs (CSRD/ESRS E1, EUDR)	annual + on request	days are acceptable
Monitoring subscriptions	weekly	24–48 h
Event alerts (wildfire, flood)	per event	max 3 h (near-real-time)

5. Volumes

Analysis is on demand per site coordinate via STAC/APIs — no bulk archive is required. Initial volume is hundreds of scenes per month and scales with customers. Cloud-native access (Copernicus Data Space Ecosystem, Element84 Earth Search, Microsoft Planetary Computer) is preferred.

6. What we ask the Luxembourg Space Agency to help with

1. Guidance on CDSE integration and on any national data-access support for early-stage companies (vouchers, ESA schemes).
2. An access route for the commercial VHR items (P7–P8) — ESA Third Party Missions, contributing missions, or vouchers — for paid validation cases only.

3. A validation partner (LIST, SnT) to raise the flood and wildfire layers from screening-level to validated.
4. Programme fit: ESA BIC Luxembourg or LuxIMPULSE — and whether a pre-company project can apply before incorporation.

7. Honest status

Our assessments are screening-level today; every value the platform produces carries its source, reference date and evidence status, and "unavailable" is stated rather than filled in. The requirements listed above are exactly what raises the platform to a validated, commercial tier — that is the purpose of this document.